

TrashTag

If you can't take it, TAG IT!



Prototype Presentation

CS 410 Fall 2025 Team Iron

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Team Bio



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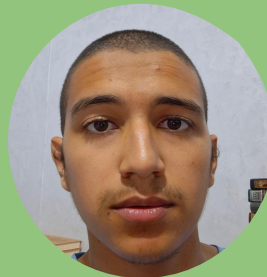
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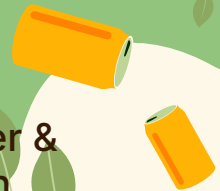
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Problem Statement



The communication gap that keeps trash on the map!



Independent organizations and local governments struggle to efficiently locate and remove litter in outdoor recreational areas such as parks, rivers, lakes, and beaches. Without knowing locations and characteristics of waste items, volunteer efforts can be left unprepared and ineffective. Communities lack a live reporting system to connect citizens with organizations capable of safe and efficient waste removal efforts.

Problem Characteristics

Identifying the Critical Gaps



01

Resources

- Organizations struggle to track illegal dumping.
- Items too much for individuals.
- Users and Litterers alike may or may not have adequate knowledge of proper disposal options.

02

Communication

- Users don't know who to contact regarding trash.
- Users lack tools focused on organizing cleanup efforts.
- User reports come in one at a time and require coordination within cleanup organizations to increase efficiency.

03

Convenience

- Users can't communicate in the moment when they see the trash.
- They've forgotten by the time they return home.
- Litterers often dump out of convenience on private or public property

04

Complexity

- Users reach out to orgs via email/social media which may get lost or go unnoticed.
- Thus, users contact multiple orgs at a time to ensure attention



Solution Statement



TrashTag aims to improve the litter and waste cleanup process by developing a mobile application where users can photograph and report piles of waste, give their exact locations, and connect with cleanup groups that are well-equipped to handle the cleanup process in order to improve environmental wellbeing and preserve nature's beauty for the public eye.

We are committed to helping the environment and the public by providing a resource that clearly highlights the location, extent, type and weight of the waste as well as provides updates regarding existing cleanup efforts, dangers and geographic littering patterns to help cleanup crews allocate their time and resources more effectively.



Solution Characteristics



Live Reporting and Mapping:

- Users snap a picture of the litter and geotag it in the app for it to be uploaded to an interactive map.
- This allows organizations to properly allocate time, resources, staff, and funding.



"Litter Hero!"

- A reward system for users who successfully clear litter locations from the map.
- This allows users track their progress and notice measurable impact.

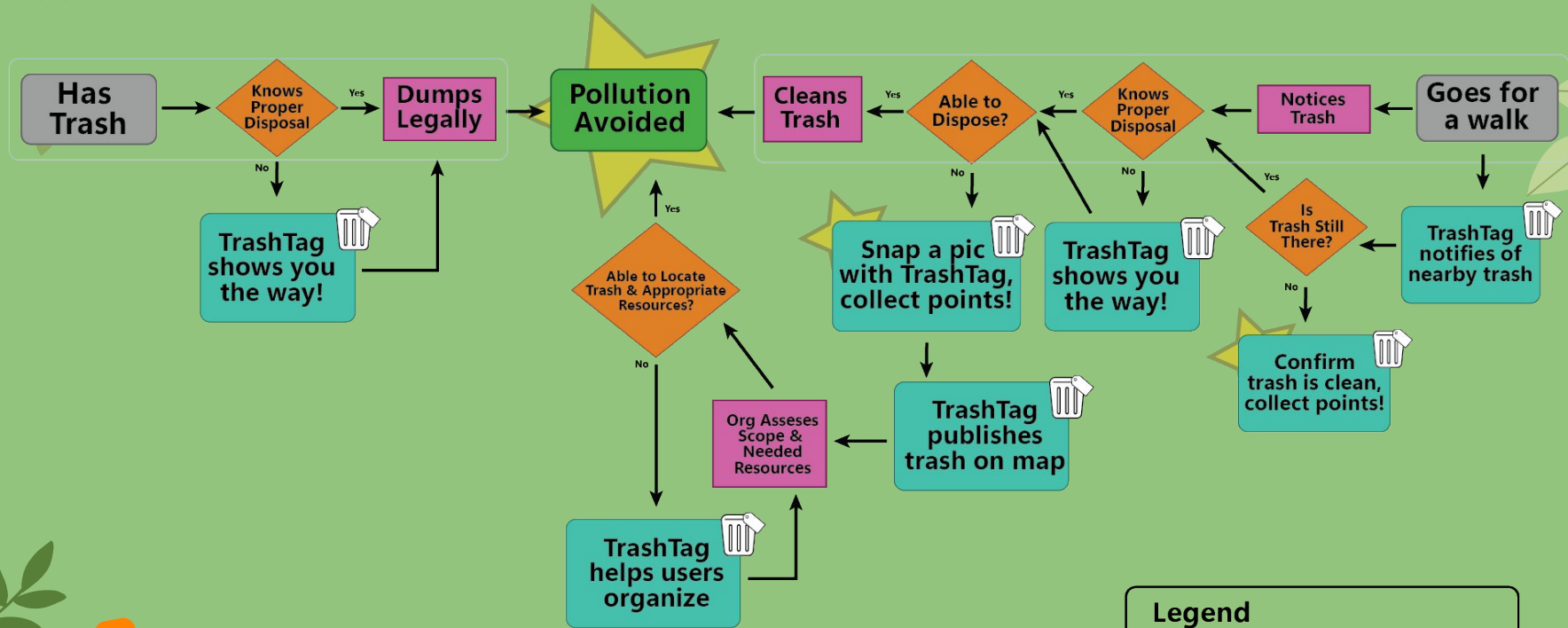
Litter Report System

- The images provide users with a visual understanding of the problem.
- This allows them to prepare resources for proper clean up.

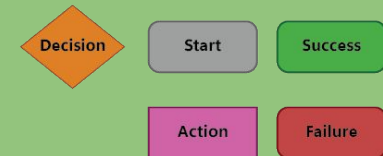
Close the loop

- When users are near reported trash they receive a request to verify the trash is still there or if its been previously cleared.

Solution Process Flow



Legend



Feature Table(1 of 6)

Category	Features	Real World	Prototype
Litter Reporting	Report Litter Location	✓	✓
	Upload Photos	✓	✓
	Report Litter Type	✓	✓
	Mark Trash Size (lbs)	✓	✓
	Request Cleanup Verification status	✓	✓
Mapping	View Litter Map	✓	✓
	Litter Density Map	✓	X - requires too much time
	Filter by Status	✓	X - nonessential
	Filter by Type	✓	X - nonessential
	Filter by Location	✓	✓



✓ - Full Implement
~ - Partial Implement
X - Eliminated



Feature Table(2 of 6)



Category	Features	Real World	Prototype
Event Coordination	Coordinate Cleanup Events	✓	X - non essential
	Claim Cleanup Area	✓	X - non essential
	Delegate Cleanup Tasks	✓	X - non essential
Event Participation	View Cleanup Events	✓	X - non essential
	RSVP to Events	✓	X - non essential
	Check-In to Events	✓	X - non essential
	Proximity Cleanup Alerts	✓	X - non essential

✓ - Full Implement
~ - Partial Implement
X - Eliminated



Feature Table(3 of 6)

Category	Features	Real World	Prototype
Cleanup Tracking	Update Cleanup Status	✓	✓
	Upload Cleanup Proof	✓	X - non essential
	Monitor Cleanup Status	✓	✓
Verification	Verify Accurate Cleanup Status	✓	✓
	Verify Cleanup Completion	✓	✓
	Flag Suspicious Content	✓	X - non essential

✓ - Full Implement
~ - Partial Implement
X - Eliminated

Feature Table(4 of 6)



Category	Features	Real World	Prototype
Community Impact	View Impact Statistics	✓	~ Only show basic stats
	Gamifications/Rewards	✓	~ Only show basic stats
	Top Contributors	✓	X - non essential
Accessible Resources	Community Guidelines	✓	X - non essential
Map Embedding	Embed Map Widget	✓	~ Demonstrate proof of concept



✓ - Full Implement
~ - Partial Implement
X - Eliminated

Feature Table(5 of 6)

Category	Features	Real World	Prototype
Analytics	Highlight Predicted Litter Trends	✓	X - non essential
	Download Reports	✓	X - non essential
	View Density - Heavy Litter Analytics	✓	X - non essential
	Track Event Impact	✓	X - non essential
	View Cleanup Statistics	✓	X - non essential

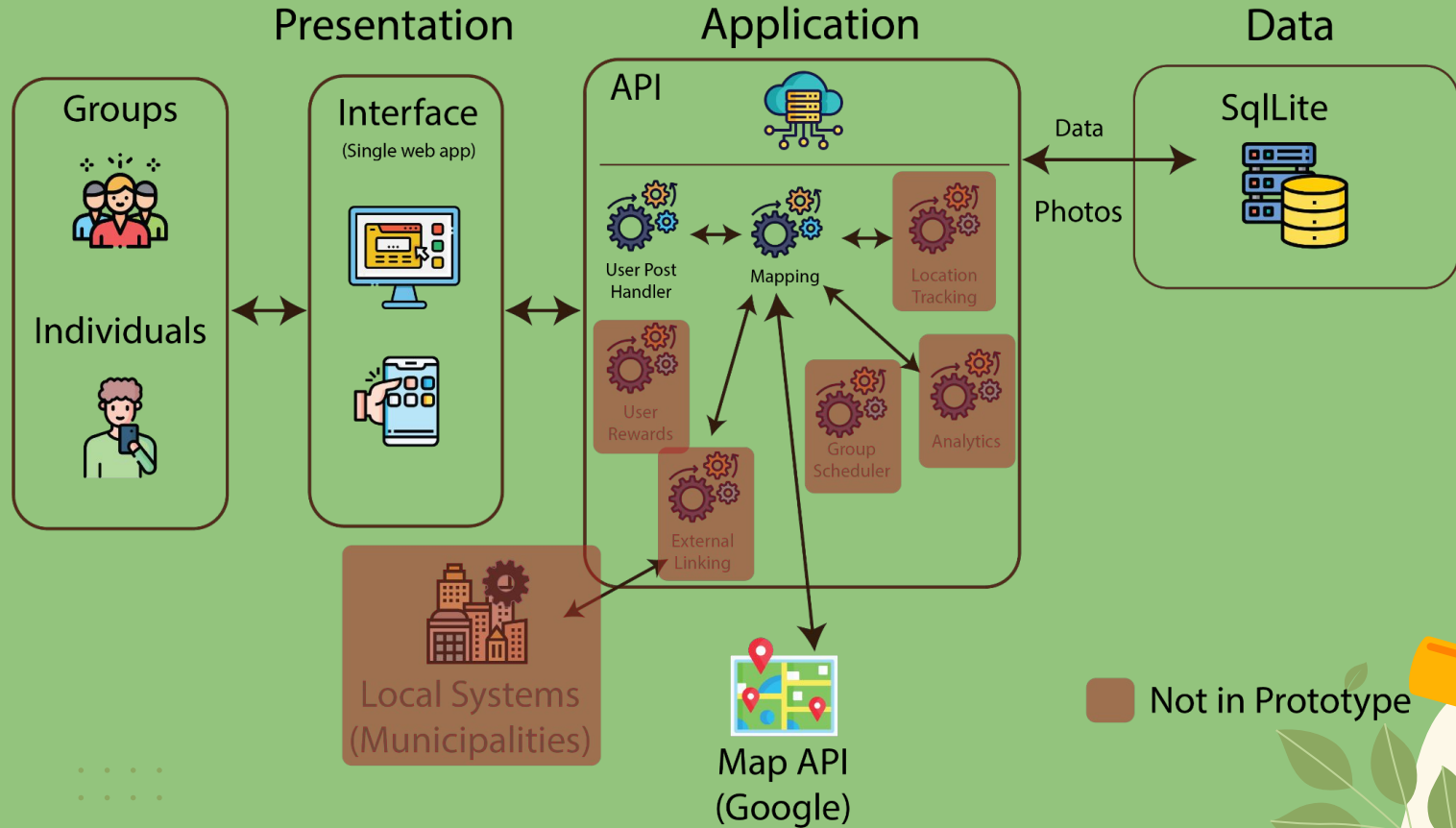
✓ - Full Implement
~ - Partial Implement
X - Eliminated

Feature Table(6 of 6)

Category	Features	Real World	Prototype
Platform Moderation	Review Flagged Reports	✓	X - non essential
	Respond to Flagged Reports	✓	X - non essential
	Edit/Correct Flagged Report Data	✓	X - non essential
	Remove Suspicious Reports	✓	X - non essential
	Mark Reports as Spam	✓	X - non essential
	Issue Violation Warnings	✓	X - non essential

✓ - Full Implement
~ - Partial Implement
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Major Functional Component Diagram



Development Tools



Development Environment	VS Code/PyCharm
Version Control	Git, GitHub
Continuous Integration and Deployment	GitHub
Frontend Languages	Javascript → React
Backend Languages	C# → Python/Django
Database	MySQL → Django ORM/SQLite
Cloud	Azure → Digital Ocean

Sprint Breakdown

Sprint 1

Setup Environments

- Github
- Setup cloud hosting
- Establishing Django ORM models

Sprint 2

Basic Client

- Show Google Map
- Capture from camera
- Verify location data

Basic API

- Image upload endpoint
- Verify location data
- Test Google API

Sprint 3

Client-API interaction

- API save data to DB
- Client upload to API
- API map data endpoint
- Client gets map data points from API

Sprint 4

Client Auth

- Create account
- Setup profile

API Auth

- New account endpoint
- Save profile endpoint

Sprint 5

Client Auth

- Edit Profile
- Login

API Auth

- Handle login
- All endpoints enforce auth

Sprint 6

- Location-based searching in background
- Notifications
- Show trash details
- Verify trash report/clean

User Risks & Mitigation



Impact	Likelihood				
	Very Low	Low	Medium	High	Very High
			R2		
		R1		R3	
		M2			
	M1		M3		

R1: False entries entered that are not trash.

R2: Minimal/no clean-up organizations in the area to handle tagged trash.

R3: Minimal/no users installing or using the app.

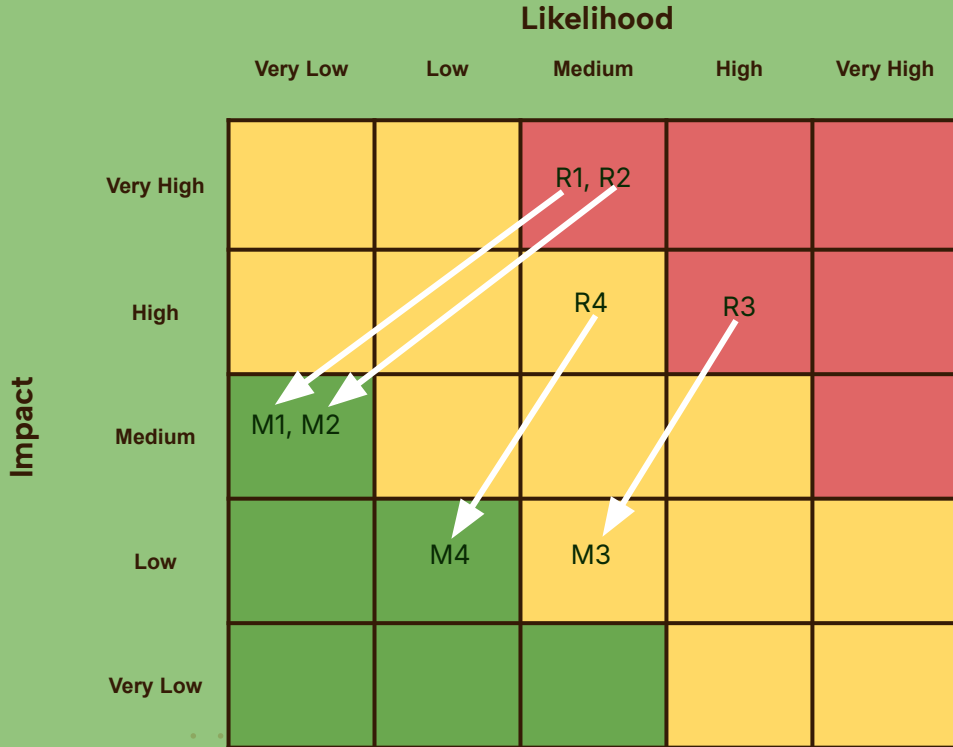
M1: Entries are verified by other users with location based notifications.

M2: Users can organize themselves. Users near tagged items receive requests to join ad hoc groups.

M3: Organization outreach/advertising



Customer Risks & Mitigation



R1: Does not integrate well with existing city/org workflows causing limited adoption

R2: Misallocation of crews due to bad/duplicate pins hurting budgets and staff morale

R3: Difficulties proving impact to local officials and donors.

R4: Budget surprises due to usage spikes and/or excessive storage needs

M1: Pilot by locale; API bridges for tailored implementation; Limit categories to not exceed capabilities

M2: Validate before dispatch recommendation; Confidence scoring; User cooldowns for inaccurate reporting

M3: Before/After photos; Outcome dashboard, Service Level Agreement (SLA) tracking

M4: Spending dashboard; Annual Total Cost of Ownership (TCO) quotes; Spending lock on non-critical features; Usage caps



Technical Risks & Mitigation



		Likelihood				
		Very Low	Low	Medium	High	Very High
Impact	Very High		R4	R1	R5	
	High		M3			R3
	Medium			R6		
	Low	M4	M1,M5			R2
	Very Low	M6		M2		

R1: Servers are down

R2: User refuses to share location

R3: Wifi connection is lost

R4: App deletes trash that still exists

R5: App presents trash that no longer exists

R6: Individual accounts are hacked

M1: Utilize reliable hosting, a backup DNS, uptime monitoring and data backups.

M2: Clarify their location is private data that will not be shared & allow choosing location on map.

M3: Add offline data persistence & auto-sync: save posts locally & publish when connection returns.

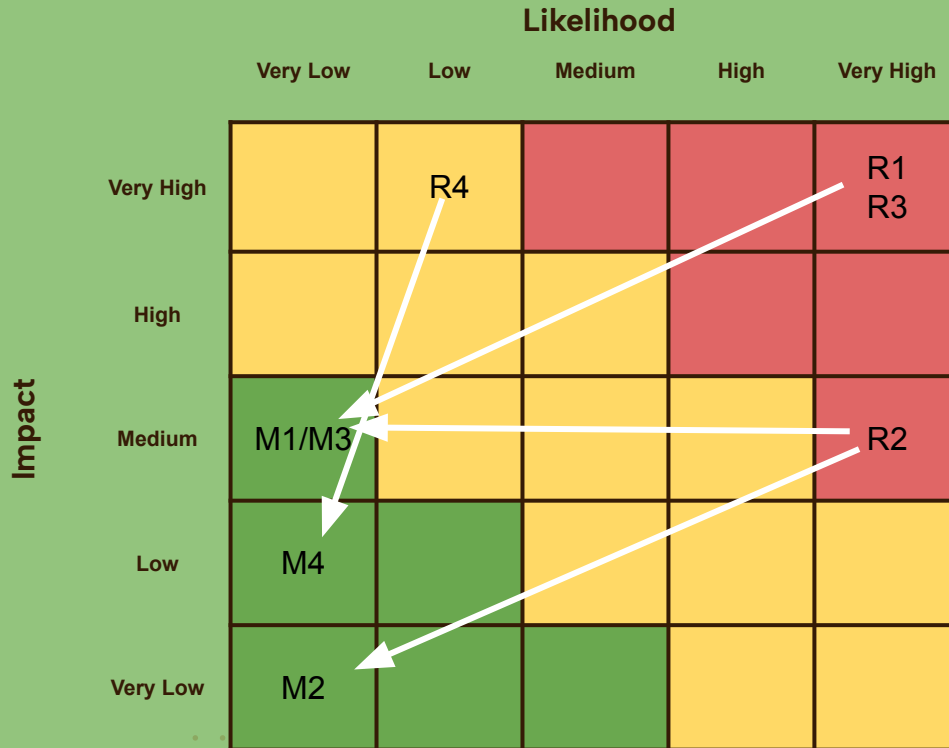
M4: Trash report not deleted until cleaner confirms.

M5: Nearby users are asked to update trash's status. Once enough users report it's removed, it is no longer displayed in the app.

M6: Use best practices for cybersecurity & reporting



Legal & Security Risks & Mitigation



R1: Users may trespass onto private property during cleanups

R2: Users may use the platform to organize crime

R3: Users may get hurt during the cleanup

R4: A data leak would likely reveal a user's location

M1: Users will have to agree to an End-User License Agreement (EULA) to use the platform that makes clear that the platform does not assume any responsibility for any injuries or arrests obtained through use of the platform

M2: Users who violates the EULA will be banned from the platform permanently

M3/M4: User data will be encrypted to avoid data leaks revealing the location of users

Conclusion



- Littering is a continuous issue prevalent in our society that affects our environment, governments, and communities.
- **The problem:** Due to the immense communication gap between citizens who see litter and cleanup organizations equipped to remove it, these organizations waste more time searching for litter rather than investing that time removing it.
- **Our solution:** A live reporting platform that closes that gap and facilitates communication through a live mapping system.
- **Engagement:** Motivating communities to “see it, report it, and clean it” through positive engagement strategies and healthy competition
- **Our vision:** By empowering communities to report litter locations, we can successfully close that gap, allowing cleanup organizations to spend more time actively removing litter rather than wasting time searching for it. This platform will transform civic duty into an engaging community movement with our powerful platform components.



Questions?



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Appendix



Application Programming Interface (API): A set of rules that allows different software programs to communicate with each other and share data or functionality.

RDBMS (Relational Database Management System): A database system that stores data in tables and organizes relationships between them.

Metadata: Data that describes other data, such as when a file was created, who created it, or what type it is.

Geotagging: The process of attaching location information to digital content, so it can be linked to a specific place on a map.

Analytics: The reports we send to local government authorities and/or organizations providing them our collected data regarding littering.

Report/Reporting: The reports that users individually enter into the app to indicate illegally dumped trash they have found in the public.

